H-B-751C
January 18, 1973
SUPERSEDING
Fed. Spec. H-B-751B
January 21, 1965

FEDERAL SPECIFICATION

BRUSH, WINDOW

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers round and oblong windown brushes, without handles.

1.2 Classification. Unless otherwise specified, the window brushes covered by this specification will be two types.

Type I - Round
Type II - Oblong

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

H-S-951	-	Synthetic Filaments For Brushes; General Specification for.
L-P-394	-	Plastic Molding Material, (Propylene Plastic, Injection and Extrusion).
R-N-91	-	Napthalene, Technical.
NN-H-104	-	Handles, Wood, Acme Threaded and Circular Tapered End.
PPP-B-1729	-	Brushes, Packaging and Packing of.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Service Centers at the General Services Administration Regional Office in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-W-6110 - Wood, Determining of Moisture Content of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of solicitation for offers or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:07Z
Check the source to verify that this is the current version before use.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 S. Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Wood. Wood for the brush blocks shall be close grained and straight grained hardwood such as beech, birch or maple. The wood shall be thoroughly and uniformly dried to a moisture content not exceeding 12 percent at the time of assembly. Moisture content shall be determined in accordance with paragraph 4.2.3.1.

3.1.2 Polypropylene. Polypropylene for the brush blocks shall be type II, grade B of L-P-394.

3.1.3 Wire.

3.1.3.1 Drawing wire. Drawing wire shall be brass, or a commercial grade of corrosion resisting steel wire, of not less than 0.015 inch diameter.

3.1.3.2 Staple wire. Staple wire shall be commercial steel staple, liquor finished, plated or coated to resist corrosion or a commercial grade of corrosion resisting steel wire. Diameter of the wire shall be not less than 0.034 inch.

3.1.4 Setting compound. The setting compound shall be brush makers pitch, epoxy resin, or other suitable setting materials which shall not be affected by the materials with which the brushes are normally used.

3.1.5 Horsetail hair. Horsetail hair shall be undyed straight, resilient, extra stiff, black or gray colored, natural horsetail hair which has been cleaned and sterilized as specified in 3.4. When tested in accordance with 4.3.3, the horsetail hair shall not perceptibly change in stiffness or show evidence of artificial coloring.

3.2 Design. The brushes shall be of the general design shown in figures 1 and 2. The illustrations are included for convenience in identifying each brush type and are not intended to preclude designs otherwise meeting the requirements of this specification.

3.3 Construction. The brushes shall be similar to types I and II as shown in figures 1 and 2 and shall conform to the requirements specified herein and in table I. The horsetail hair specified in 3.1.5 shall be set as specified in 3.3.2, into blocks specified in 3.3.1, so that when tested as specified in 4.3.1, the tufts shall not separate from the blocks. The horsetail hair weight shall be not less than shown in table I, when tested as specified in 4.3.2. The brushes shall be trimmed to a square top.

3.3.1 Blocks. Blocks shall be constructed of wood specified in 3.1.1 or polypropylene specified in 3.1.2 at the option of the manufacturer, and shall conform to the dimensions in table I. The brush side of the block shall have holes drilled as specified in table I to receive and uniformly distribute the horsetail hair tufts. The handle side of the block shall have tapered holes specified in 3.3.1.3, to receive the handles. All corners shall be broken, all edges shall be rounded. All surfaces shall be smoothly finished and free of splinters or torn or loose grain.

3.3.1.1 Type I. The type I block shall be round and shall have two tapered handle holes as specified in 3.3.1.3, approximately 18 degrees from the face of the brush diametrically opposed to each other.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:07Z
Check the source to verify that this is the current version before use.

3.3.1.2 Type II. The type II blocks shall be oblong and shall have two tapered handle holes, as specified in 3.3.1.3. The holes shall be centered on the long side of the block, approximately 45 degrees from the brush face of the block, directly opposite from each other.

3.3.1.3 Holes for handles. The holes for handles shall be compatible with the requirements for type II, size I, handles specified in NN-H-104. Handle holes shall have a minimum depth of not less than 1 inch.

TABLE I. Physical requirements

	Type I Round	Type II Oblong
Block		
Diameter, inches	4 1/2 ± 1/4	
Thickness, inches	1 3/4 ± 1/8	
Turned out at top width, inches	3 1/8 ± 1/4	
Length, inches		8 ± 1/4
Width, inches		2 3/8 ± 1/8
Thickness in center, inches		1 7/8 ± 1/8
Thickness on end, inch		7/8 ± 1/8
Horsehair:		
Weight of horsehair per brush, minimum, ounces	6	4 7/32
Length clear of block, minimum inches	2 1/4	2 1/4
Holes for Horsehair:		
Number, minimum	65	92
Depth, minimum inch	5/16	5/16
Diameter, minimum, inch	7/32	7/32

3.3.2 Set. Any method described below may be used for setting the horsetail hair in the block.

3.3.2.1 Staple set. The horsetail hair shall be cut to double length and center folded in quantities to insure individual tufts of such size to fit snugly when inserted in the holes provided in the block. Stapling shall be accomplished so that the staples, made from wire specified in 3.1.3.2, and the horsetail hair are driven to the bottom of the holes with the staple securely imbedded in the wood.

3.3.2.2 Compound set. The horsetail hair shall be cut to suitable lengths for tufts. The setting compound specified in 3.1.4, shall be rubbed or worked into the butt end of the tuft, then the butt end shall be wound with five turns of thread. The butt end of the tuft shall be redipped in the setting compound and inserted in the block hole.

3.3.2.3 Wire drawn. Continuous lengths of drawing wire specified in 3.1.3.1, shall be looped up from the underside of the block in predrilled holes. Horsetail hair, cut to suitable lengths and of sufficient quantity to fill the holes provided, shall be secured at the centers in the drawn loops of wire. The drawing wire in the finished brush shall be concealed within the brush head.

3.4 Sterilization. Horsetail hair shall be sterilized at some point in the manufacturing process, after thorough washing and rinsing, in accordance with any of the procedures specified in 3.4.1, 3.4.2, or 3.4.3. A properly checked steam-pressure gauge, for the method in 3.4.3, and a thermometer, for the methods in 3.4.1 and 3.4.2, visible from the outside of the chamber, shall be provided. A suitable recording device to provide a record of the pressure and temperature, as applicable, of the disinfecting chamber shall also be provided.

3.4.1 Boiling water. Horsetail hair shall be soaked in boiling water (212°F.) for not less than 30 minutes.

3.4.2 Dry heat. Horsetail hair shall be placed in a chamber and a temperature above 284°F. shall be developed and maintained for not less than 3 hours.

3.4.3 Autoclaving. Horsetail hair shall be placed in a steam pressure chamber for a minimum of 45 minutes at not less than 15 p.s.i.g.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:07Z
Check the source to verify that this is the current version before use.

3.5 Finish.

3.5.1 Wood blocks. Wood blocks shall be coated with varnish, lacquer, or other suitable clear organic material in accordance with the manufacturer's standard commercial practice.

3.6 Marking for identification. Each block shall be permanently marked with the name of the manufacturer, or an abreviation thereof or trademark of such known character that the source of manufacture can be readily identified.

3.7 Workmanship. The brush shall conform to the quality and grade of product established by this specification. Brushes shall be free from defects which may affect their durability, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the examination and testing set forth in the specification where such are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.2.1 Inspection of components and materials. In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, examined and tested in accordance with the specified requirements of referenced subsidiary specifications and standards or if none, in accordance with this specification.

4.2.2 Process examination. Examination shall be made of the sterilization of the horsetail hair to establish that no deviation is made from the specified procedure or requirement. Whenever a deviation is noted, correction shall be made to the items affected and lot in process.

4.2.3 Intermediate testing.

4.2.3.1 Wood. Wood components shall be tested in accordance with MIL-W-6110 to determine compliance with 3.1.1 as applicable the sample unit shall be one block or handle as applicable. The lot shall be the number of blocks or handles as applicable offered for acceptance at one time. The test shall be made immediately previous to assembly and finish. Three determinations shall be taken on each sample unit and results shall be the average of the three determinations. The inspection level shall be level S-2 with an acceptable quality level (AQL) of 2.5, expressed in terms of defects per hundred units.

4.2.4 End item inspection. The lot shall be all window brushes of the same type offered for inspection at one time. The sample unit shall be one completely fabricated window brush.

4.2.4.1 Visual inspection. The end item shall be examined for the defects in table II. The inspection level shall be level II and the acceptable quality level (AQL) shall be 4.0.

TABLE II. Classification of defects

Examine	Defect
Material	
Wood Block	Not close grained and straight grained hardwood free of splinters or loose grain.
	Not coated as specified.
Polypropylene block	Not as specified.
Horsetail hair	Not clean, evidence of dirt or foreign matter.
Construction and workmanship	Not of the general design for type I or II as shown on figure 1.
	Block does not have two tapered handle holes, not directly opposite from each other. Not pitch set or equivalent, wire drawn, or staple set.
	Brush filler material not evenly distributed in brush head.
Marking for identification	Omitted, incomplete, not legible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:07Z
Check the source to verify that this is the current version before use.

4.2.4.2 Dimensional examination. Brushes shall be examined for conformance with dimensional requirements specified. The inspection level shall be S-2, and the acceptable quality level (AQL) shall be 4.0 percent defective. The unit for examination shall be one brush.

4.2.4.3 Testing of the end item. The end item shall be tested for the applicable characteristics as indicated in table III. The inspection level and AQL shall be as indicated in table III.

TABLE III. Instructions for testing

Characteristic	Specification reference		Requirements applicable to individual unit	Number determinations per unit	Results reported as pass or fail	Inspect level	AQL
	Requirement	Test method					
Tightness of horsetail hair	3.3	4.3.1	X	10	X	S-2	4.0
Weight of horsetail hair	3.3	4.3.2	X	1	X	S-2	4.0
Stiffness of horsetail hair	3.1.5	4.3.3	X	1	X	S-2	4.0
Color of horsetail hair	3.1.5	4.3.4	X	1	X	S-2	4.0

4.3 Tests.

4.3.1 Set retention. Secure the brush block in a suitable stationary clamping device and the ends of a tuft of horsetail hair in another clamp so that they are evenly distributed to prevent distortion and cutting. Care shall be taken to assure that both ends of the same hair are not held in the clamping device. By means of a spring scale or weight, apply a force of 2 ± 0.5 pounds to the tuft to determine compliance with 3.3. Ten specimens from a brush shall be tested. Failure of one specimen in a brush shall be considered a defect.

4.3.2 Horsetail hair weight.

4.3.2.1 Removal of horsetail hair. Horsetail hair set in setting compound shall have the setting compound removed from the butt end of the knotted tufts by immersing the butt end in gasoline or other suitable solvent to a depth of not more than 1/8 inch above the setting compound. After the setting compound has been removed, the solvent shall be allowed to evaporate and the thread shall be removed from the tufts.

4.3.2.2 Conditioning of horsetail hair. The horsetail hair removed from the brush as specified in 4.3.2.1 shall be conditioned for not less than 16 hours in air at a relative humidity of 50 ± 4 percent and at a temperature of 23° ± 1.1°C. Weighing to determine compliance with the weight requirements specified in 3.3 shall be performed in the same atmosphere.

4.3.3 Stiffness test. A complete brush, minus handle, shall be soaked in boiling water (212 F.) for not less than one hour. The brush shall be removed from the water and placed face side up along side a finished brush, of the same type, that has not been soaked in the boiling water. The horsetail hair in both brushes shall be depressed with the palms of the hands for not less than 10 times respectively to determine compliance with 3.1.5.

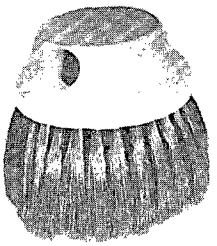
4.3.4 Color test. The horsetail hair tested as specified in 4.3.3 and the water used for the test shall be examined for compliance with 3.1.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging and packing shall be in accordance with the requirements of PPP-B-1729 as specified for the applicable levels (see 6.2).

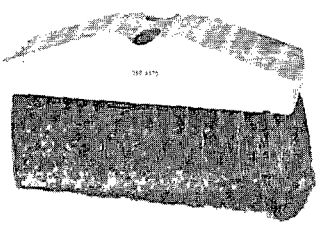
5.2 Marking. Containers of the brushes shall be marked in accordance with the requirements of PPP-B-1729.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:07Z
Check the source to verify that this is the current version before use.



TYPE I ROUND

FIGURE 1



TYPE II OBLONG

FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:07Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The brushes covered by this specification are intended to be used for washing windows.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type of brush required (see 1.2).
(c) Selection of applicable packaging and packing levels (see 5.1 and 5.2).

Preparing Activity:

GSA

User Activities:

Navy - MC, SA
Air Force - 54
Army - GL

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. **Price 10 cents each.**

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:07Z
Check the source to verify that this is the current version before use.